

Chapter 1. Some Basics

1.1 Functions

It is common that the values of one variable depend on the values of another. E.g. the area A of a region on the plane enclosed by a circle depends on the radius r of the circle ($A = \pi r^2$, $r > 0$.) Many years ago, the Swiss mathematician Euler invented the symbol $y = f(x)$ to denote the statement that “ y is a function of x ”.

A function represents a rule that assigns a *unique* value y to each value x .

We refer to x as the *independent variable* and y the *dependent* variable.

One can also think of a function as an input-output system/process: input the value x and output the value $y = f(x)$. (This becomes particularly useful when we combine or composite functions together.)

1.2 Operations on Functions

1.2.1 Arithmetical operations

Let f and g be two functions.

(i) The functions $(f \pm g)(x) = f(x) \pm g(x)$, called the sum or difference of f and g .

(ii) The function $(fg)(x) = f(x)g(x)$, called the product of f and g .

(iii) The function $(f/g)(x) = f(x)/g(x)$, called the quotient of f by g , is defined where $g(x) \neq 0$;

1.2.2 Composition

Let $f : D \rightarrow \mathbb{R}$ and $g : D' \rightarrow \mathbb{R}$ be two (real) functions with domains D and D' respectively.

The function

$$(f \circ g)(x) = f(g(x)),$$

called *f composed with g* or *f circle g*, is defined on the subset of D' for which the values $g(x)$ (i.e. the range of g) are in D .

1.2.3 Example

Let $f(x) = x - 7$ and $g(x) = x^2$ (defined on all of $\mathbb{R}$). Then

$$(f \circ g)(2) = f(g(2)) = f(4) = -3, \quad \text{and}$$

$$(g \circ f)(2) = g(f(2)) = g(-5) = 25.$$

Note that in general $f \circ g \neq g \circ f$.

1.3 Limits

In this section we are interested in the behaviour of f as x gets closer and closer to a .

1.3.1 Example

Let $D = \{x \in \mathbb{R} : x \neq 0\}$ and we consider the function $f : D \rightarrow \mathbb{R}$ given by $f(x) = \frac{\sin(x)}{x}$. (x is in radian.) Describe its behaviour as x tends to 0.

Clearly when $x = 0$, $\frac{\sin(0)}{0} = \frac{0}{0}$ does not make sense.

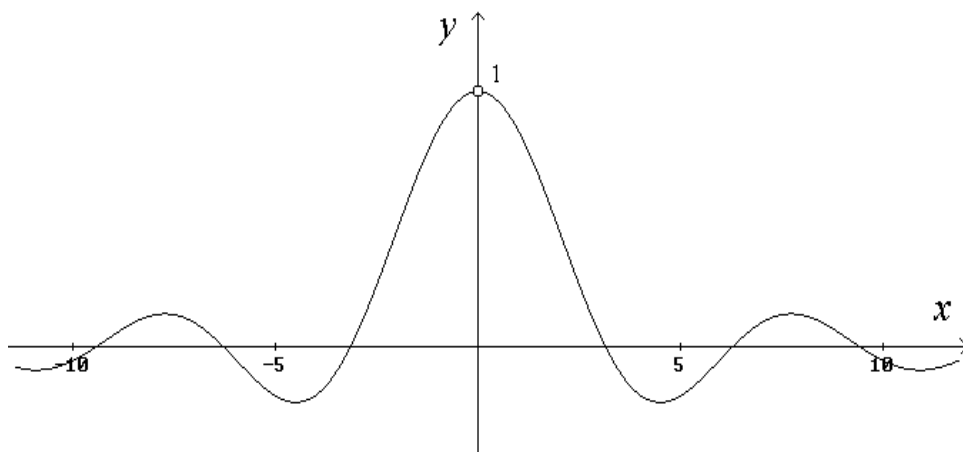
It is defined everywhere except at 0 and thus it makes sense to ask how it behaves as it is evaluated at ar-

guments which are closer and closer to 0.

If we plot the graph of $\frac{\sin(x)}{x}$, we see that as x gets closer and closer to 0 from either sides (and not reaching 0 itself), $\frac{\sin(x)}{x}$ approaches 1. In this case, we say that “the limit of $\frac{\sin(x)}{x}$ as x tends to 0 is equal to 1”.

We use the following notation:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1.$$



1.3.2 Informal Definition

Let $f(x)$ be defined on an open interval I containing x_0 , except possibly at x_0 itself. If $f(x)$ gets arbitrary close to L when x is sufficiently close to x_0 , then we say that the limit of $f(x)$ as x tends to x_0 is the number L and we write

$$\lim_{x \rightarrow x_0} f(x) = L.$$

1.3.3 Rules of Limits

Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = L'$, then the following statements are easy to verify:

- (i) $\lim_{x \rightarrow a} (f \pm g)(x) = L \pm L'$;
- (ii) $\lim_{x \rightarrow a} (fg)(x) = LL'$;

$$(iii) \lim_{x \rightarrow a} \frac{f}{g}(x) = \frac{L}{L'} \text{ provided } L' \neq 0;$$

$$(iv) \lim_{x \rightarrow a} kf(x) = kL \text{ for any real number } k.$$

1.3.4 Example.

Consider how to evaluate

$$\lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1}.$$

We cannot substitute $x = -1$, and the numerator and denominator have no obvious common factors.

The idea is to multiple the numerator and denominator by the *conjugate expression* $\sqrt{x^2 + 8} + 3$:

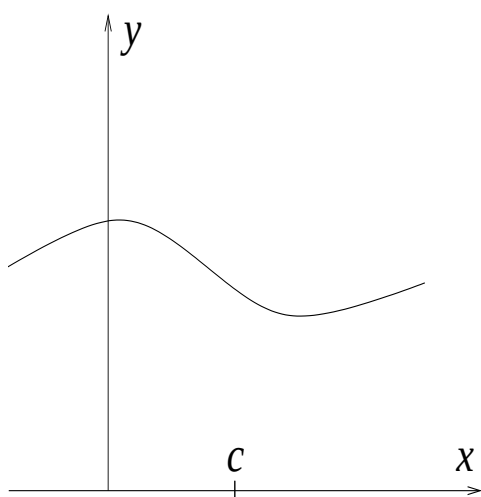
$$\begin{aligned} \frac{\sqrt{x^2 + 8} - 3}{x + 1} &= \frac{(\sqrt{x^2 + 8} - 3)(\sqrt{x^2 + 8} + 3)}{(x + 1)(\sqrt{x^2 + 8} + 3)} \\ &= \frac{(x + 1)(x - 1)}{(x + 1)(\sqrt{x^2 + 8} + 3)} = \frac{x - 1}{(\sqrt{x^2 + 8} + 3)}. \end{aligned}$$

The cancelation occurs **when** $x \neq -1$. Hence

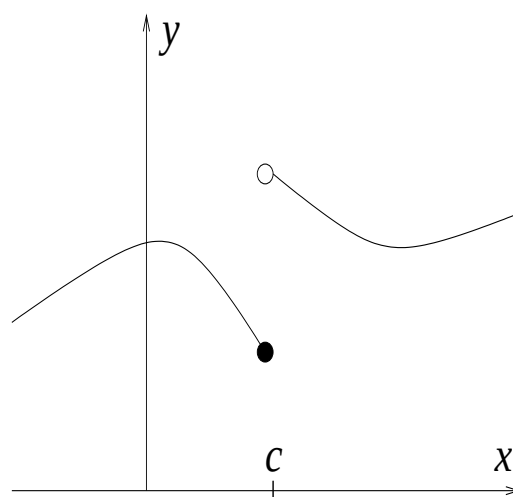
$$\lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1} = \lim_{x \rightarrow -1} \frac{x - 1}{(\sqrt{x^2 + 8} + 3)} = -\frac{1}{3}.$$

1.4 Continuity

Continuity is an intuitive concept. Intuitively, a function is continuous if we can draw its graph “in one stroke”, or “without lifting up the pen from the paper”.



Continuous at c



Not continuous at c

1.4.1 Example

$f(x) = 1/x$ is continuous at every x except $x = 0$

where it is not defined.

1.4.2 Example

We accept without proof the following two facts:

1. All polynomials are continuous at every point in

$\mathbb{R}$.

2. All rational functions (quotient of 2 polynomials)

$p(x)/q(x)$ where p and q are polynomials are contin-

uous at every point such that $q(x) \neq 0$.

1.4.3 Continuous functions

A function $f : D \rightarrow \mathbb{R}$ is called a *continuous* func-

tion if f is continuous at ALL points in D .